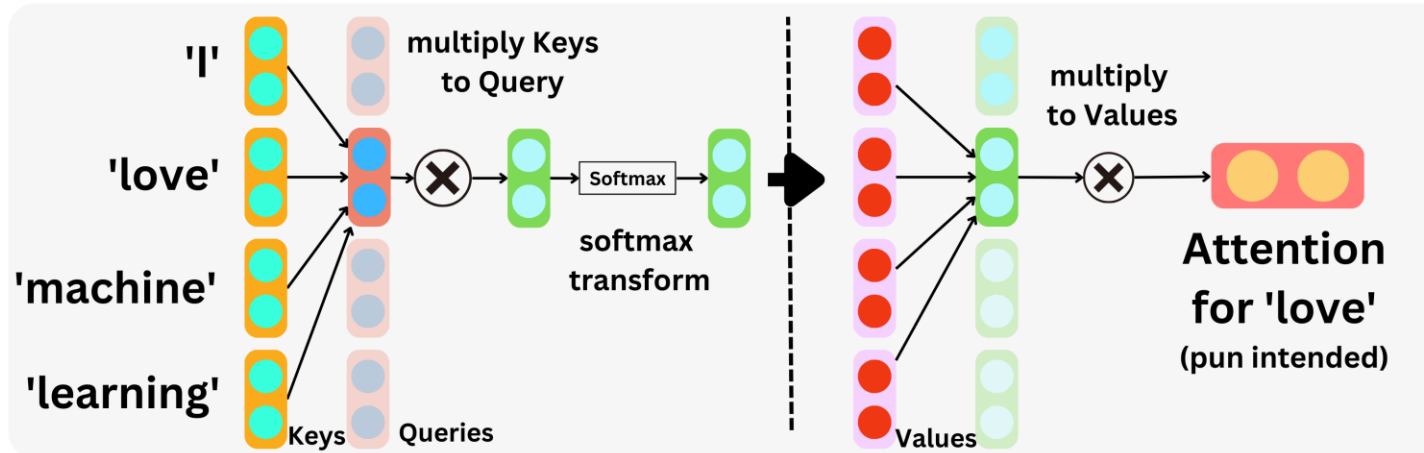
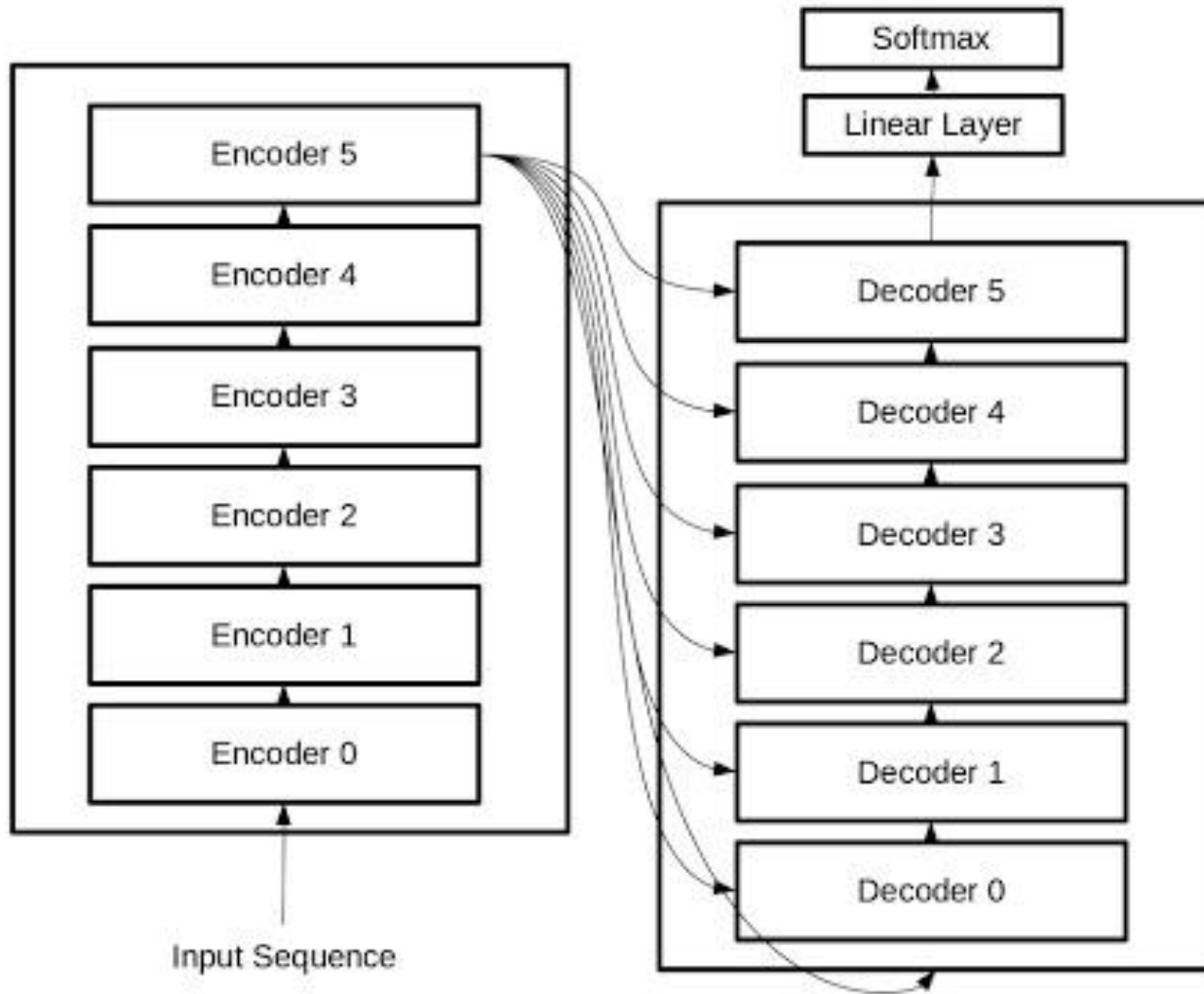


Step 2: Create the Attention



Self-Attention: Understanding Context

- Self-attention allows a model to "look" at all words in a sentence simultaneously to understand the context of a specific word.
- It creates a dynamic relationship between words regardless of how far apart they are in a sentence.
- The mechanism uses three distinct vectors for each word: **Query (Q)**, **Key (K)**, and **Value (V)**.



The Transformer Block: Deep Dive into Architecture

- The Transformer architecture is built on an **Encoder-Decoder** structure.
- **Encoder**: Processes the input sequence and creates a mathematical representation of it.
- **Decoder**: Uses the encoder's output and previous generated tokens to predict the next word.
- This architecture is the foundation for models like GPT, Llama, and DeepSeek.

Multi-Head Attention:

Focusing on Multiple Details

- Instead of performing attention once, "Multi-Head" attention runs multiple attention mechanisms in parallel.
- This allows the model to focus on different parts of a sentence simultaneously (e.g., one head focuses on the subject, another on the verb).
- The results from all "heads" are concatenated and transformed to produce the final output.

